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FOR FINAL TERM (FALL 2021,22)

Course Code: MTH631

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SYED HUSSNAIN ALI NAQVI

MSC MATHEMATICS

Question No : 1 of 38

The graph of an odd function is symmetrical about ---------------

Answer (Please select your correct option)

- x-axis
- y-axis
- origin
- None of these

even function y axis
odd function origion

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Question Summary : (Attempted Question)

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If $h = f(x, y), x = f(r, \theta)$ and $y = f(r, \theta)$, then using the chain rule, $\frac{\partial h}{\partial \theta} = \text{.........}$

Answer (Please select your correct option)

- $\frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial \theta} + \frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial \theta}$
- $\frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial \theta}$
- $\frac{\partial f}{\partial x} + \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} + \frac{\partial y}{\partial \theta}$
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 14 of 34

If $h = f(x,y), x = r\cos\theta$ and $y = r\sin\theta$, where then using the chain rule, $\frac{\partial h}{\partial r} =$ ------

Answer (Please select your correct option)

$\cos\theta\frac{\partial f}{\partial\theta} + \sin\theta\frac{\partial f}{\partial r}$

$\cos\theta\frac{\partial f}{\partial x} + \sin\theta\frac{\partial f}{\partial y}$

Hussnain

$\cos\theta\frac{\partial f}{\partial x} - \sin\theta\frac{\partial f}{\partial y}$

None of these

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Question Summary : (Attempted Question)

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Question No :

_____________ is the only region in R.

Answer (Please select your correct option)

- Set of finite points of R
- Interval
- Any finite sequence of R
- None of these

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Question Summary : (Attempted Question)

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For the function $f(x,y)=\frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y)\to(0,0)$ along the line $y=x$ is ----

Answer (Please select your correct option)

- $\frac{1}{2}$ p 87
- $-\frac{1}{2}$
- 0
- 1

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MTH631 Real Analysis II

Question No : 2 of 34

If H is an open covering of a compact subset S then, S can be covered by ----------

Answer (Please select your correct option)

- infinitely many sets from H
- finitely many sets from H p 79
- atleast one set from H
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 1 of 26

The singleton set is always —

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Answer (Please select your correct option)

- disconnected
- polygonally disconnected
- connected
- None of these

ATH631 Real Analysis II

Question No : 2 of 26

Any polygonally connected set either it is open or not, is always ________

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Answer (Please select your correct option)

- connected
- disconnected
- a region
- None of these

MTH631 Real Analysis II

Question No : 5 of 26

Marks: 1 (Budgeted Ti

Suppose $U_0 \in T$ is a limit point of T, g is continuous at U_0 and f is continuous at $X_0 = G(U_0)$. Then ——— is continuous at U_0.

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Answer (Please select your correct option)

- $f \circ g$ p 94
- $\frac{f}{g}$
- $\frac{g}{f}$
- None of these

MTH631 Real Analysis II

Question No : 7 of 26 Marks: 1 (Bud

A function f is ———— on a subset S of R^n if S is contained in an open set on which $f_{x_1}, f_{x_2}, \dots f_{x_n}$ are continuous

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Answer (Please select your correct option)

- continuously differentiable
- uniformly convergent
- piecewise convergent
- None of these

MTH631 Real Analysis II

Question No : 18 of 26

A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \dots + m_n x_n$ for some constants m_i's is called ——

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Answer (Please select your correct option)

- linear function
- quadratic function
- constant function
- None of these

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MTH631 Real Analysis II

Question No : 20 of 34

If f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|dx < \infty$, then $\int_a^b f(x)dx$ is ----------

Answer (Please select your correct option)

- divergent
- conditionally convergent
- convergent ✓ b.pg 138
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 21 of 34

Marks: 1 (Budgeted Time Min)

Suppose f is a continuous and $F(x)=\int_a^x f(t)dt$ is bounded on $[a,b)$. Let g' be an absolutely integrable on $[a,b)$ and suppose $\lim_{x\to b^-} g(x)=0$

Then----------

Answer (Please select your correct option)

- $\int_a^b f(x)g(x)dx$ converges ✓ b.pg 139
- $\int_a^b f(x)g(x)dx$ diverges
- $\int_a^b f(x)g(x)dx$ conditionally converges
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 18 of 34 Marks: 1

A function f is said to be absolutely integrable on $[a,b)$ if f is locally integrable on $[a,b)$ and -------------

Answer (Please select your correct option)

- $\int_a^b |f(x)|dx < \infty$ ✓

b.pg 137

- $\int_a^b |f(x)|dx = \infty$
- $\int_a^b |f(x)|dx \geq \infty$
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 24 of 34

The lower integral $\underline{\int_R} f(X)d(X)$ of f over R is the supremum of all the ----------.

Answer (Please select your correct option)

- lower sums
- upper sums
- average sums
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 19 of 34

An absolutely convergent integral is always ----------------

Answer (Please select your correct option)

- divergent
- convergent
- conditionally convergent
- None of these

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19

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 23 of 34

The upper integral $\overline{\int_R} f(X)d(X)$ of f over R is the infimum of all the-------------

Answer (Please select your correct option)

- lower sums
- upper sums ✓
- average sums
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 15 of 34

If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \le 0$ for all $X \ne X_0$. Then p is called ______

Answer (Please select your correct option)

- positive semidefinite
- positive definite
- negative semidefinite ✓
- negative definite

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Question Summary : (Attempted Question)

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Question No : [illegible]

If f is unbounded on the nondegenerate rectangle R in $\mathbb{R}^n$, then______

Answer (Please select your correct option)

- f is not integrable on R ✓

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- f is integrable on R
- f is not differentiable on R
- None of these

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Question Summary : (Attempted Question)

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Question No : 12 of 34

A function of the form $L(X) = f_{x_1}(X_0)x_1 + f_{x_2}(X_0)x_2 + \ldots + f_{x_n}(X_0)x_n$ is called ———

Answer (Please select your correct option)

- quadractic function
- differential of f at X_0
- constant function
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 25 of 34

If f is integrable on a rectangle R, then --------

Answer (Please select your correct option)

- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X) = \int_R f(X)d(X)$
- $\overline{\int_R} f(X)d(X) = \underline{\int_R} f(X)d(X) = \int_R f(X)d(X)$ 150
- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X) > \int_R f(X)d(X)$
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 9 of 34

A point on the graph of a function at which the tangent line is parallel to x-axis is called the ....

Answer (Please select your correct option)

- critical point
- point of inflection
- point of relative extrema
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 10 of 34

The rate of change of $f(x,y)$ in the direction of a unit vector is called the ---------------- of $f(x,y)$.

Answer (Please select your correct option)

- differential
- directional derivative
- mixed derivative
- None of these

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Question Summary : (Attempted Question)

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Question No : 11 of 34

Marks:

A function of several variables may have first order partial derivatives at a point X_0 but ---------- continuous at X_0.

Answer (Please select your correct option)

- fail to be
- must be
- always not
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 28 of 34

A ---------- surface in $\mathbb{R}^n$ has zero content.

Answer (Please select your correct option)

- parabolic
- elliptical
- differentiable
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 27 of 34

Any bounded set E with only finitely many ---------------- has zero content.

Answer (Please select your correct option)

- limit points
- neighbourhood points
- interior points
- None of these

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Question Summary : (Attempted Question)

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A set of points of the form $X = X_0 + tU,\ t_1 \le t \le t_2$ is called ......

Answer (Please select your correct option)

- the neighbourhood.
- line segment.
- the closure.
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 5 of 34

If f and g are continuous functions at the point X_0 in R^n, then $\frac{f}{g}$ is continuous at X_0 provided that ——

Answer (Please select your correct option)

- $g(X_0)=0$
- $g(X_0)\neq 0$ ✓
- $f(X_0)-g(X_0)=0$
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 29 of 34

If f is integrable on disjoint sets S_1 and S_2 then f is integrable on

Answer (Please select your correct option)

- $S_1 \cup S_2$
- S_1 / S_2
- S_2 / S_1
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 28 of 34

A ------------ surface in $\mathbb{R}^n$ has zero content.

Answer (Please select your correct option)

parabolic

elliptical

differentiable

None of these

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MTH631 Real Analysis II

Question No : 27 of 34

The ------------ of finitely manay sets with zero content has zero content.

Answer (Please select your correct option)

- complement
- uninion
- diference
- None of these

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MTH631 Real Analysis II

Question No : 14 of 31

If value of a becomes zero in the Taylor series $\sum_{n=0}^{\infty}\frac{f^{n}(a)}{n!}(x-a)^{n}$, then the series is called ------

Answer (Please select your correct option)

- Maclaurin series
- Fourier series
- Power series
- None of these

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MTH631 Real Analysis II

Question No : 30 of 34

A set is closed if and only if it contains all of its ———

Answer (Please select your correct option)

- Neighborhoods
- Limit points
- Exteriors
- None of these

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MTH631 Real Analysis II

Question No : 20 of 25

If f and g are integrable on S and $f(X) \le g(X)$ for X in S, then ———

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Answer (Please select your correct option)

- $\int_S f(X)\,dX \ge \int_S g(X)\,dX$
- $\int_S f(X)\,dX \le \int_S g(X)\,dX$
- $\int_S f(X)\,dX > \int_S g(X)\,dX$

b is correct page 155

- None of these

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MTH631 Real Analysis II

Question No : 15 of 26

The function $f(x) = (1-x)^{-p}$ is locally integrable on ———

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Answer (Please select your correct option)

- $[0,1]$
- $[-1,1)$
- $(0,1)$
- $[0,1)$

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MTH631 Real Analysis II

Question No : 11 of 26

If $f(x) = 3x + 5x^3$, then at $x = -1$, $df =$ ______

Answer (Please select your correct option)

- $18dy$
- $18dx$
- 18

3+15x^2

3+15(-1)^2

3+15=18

MTH631 Real Analysis II

Question No : 7 of 26

A $f_x, f_x, ... f_x$ exist on a neighborhood of X_0 and are continuous at X_0, then f is ——— at X_0

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Answer (Please select your correct option)

- uniformly converget
- piecewise convergent
- differentiable
- None of these

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MTH631 Real Analysis II

Question No : 6 of 26

A set A □ R of real numbers is ————— if there exists a real number M □ R, such that x ≤ M for every x □ A.

Answer (Please select your correct option)

- bounded below
- uniformly continuous
- bounded above
- None of these

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MTH631 Real Analysis II

Question No : 5 of 26

In a vector valued function $G = (g_1, g_2, \ldots, g_n)$. Then $g_1, g_2, \ldots, g_n$ are called the ———— functions of G .

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Answer (Please select your correct option)

- bijective
- surjective
- component
- None of these

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MTH631 Real Analysis II

Question No : 4 of 26

The quotient of two continuous functions is a ———— function wherever the denominator is non-zero.

Answer (Please select your correct option)

- differentiable
- continuous
- composite
- None of these

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MTH631 Real Analysis II

Question No : 3 of 26

The set $S = \{(x,y), x^2 + y^2 \leq 1 \text{ or } x^2 + y^2 \geq 4\}$ is not a region in R^2, since it is ——————

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Answer (Please select your correct option)

- not connected
- connected
- polygonally connected
- None of these

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MTH631 Real Analysis II

Question No : 2 of 26

A set S is polygonally connected if every pair of points in S can be connected by a polygonal path lying ———

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Answer (Please select your correct option)

- outside the set S
- partially in S
- entirely in S
- None of these

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MTH631 Real Analysis II

Question No : 1 of 26

A set is closed if and only if it always contains all of its ————

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Answer (Please select your correct option)

- limit points
- neighbourhood points
- interior points
- None of these

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Question No : 20 of 26

A ———— surface in $\mathbb{R}^n$ has zero content.

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Answer (Please select your correct option)

- parabolic
- elliptical
- differentiable
- None of these

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H631 Real Analysis II

estion No : 17 of 26 Marks: 1 (Budgeted Time

f and g are locally integrable on $[a,b)$, $g(x) > 0$ and $f(x) \geq 0$ on some subintervals $[a_1,b)$ of $[a,b)$, and $\lim_{x \to b-} \frac{f(x)}{g(x)} = M$. If $M = \infty$ and $\int_a^b g(x)dx = \infty$, th

page 144

swer (Please select your correct option)

- $\int_a^b f(x)dx$ and $\int_a^b g(x)dx$ converges together
- $\int_a^b f(x)dx \geq \infty$
- $\int_a^b f(x)dx = \infty$
- $\int_a^b f(x)dx < \infty$

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Question No : 14 of 26

If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \geq 0$ for all X, then p is called ————

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Answer (Please select your correct option)

- positive semidefinite
- positive definite
- negative semidefinite
- negative definite

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MTH631 Real Analysis II

Question No : 7 of 26 **Marks: 1**

A function f is ———— on a subset S of R^n if S is contained in an open set on which $f_{x_1}, f_{x_2}, \ldots f_{x_n}$ are continuous

page 116

Answer (Please select your correct option)

- continuously differentiable
- uniformly convergent
- piecewise convergent
- None of these

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If f is bounded on a rectangle R, then ---------------

Answer (Please select your correct option)

- $\overline{\int}_R f(X)d(X) \le \underline{\int}_R f(X)d(X)$
- $\overline{\int}_R f(X)d(X) > \underline{\int}_R f(X)d(X)$
- $\overline{\int}_R f(X)d(X) \ge \underline{\int}_R f(X)d(X)$
- None of these

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A bounded function f is integrable on a rectangle R if and only if ---------

Answer (Please select your correct option)

- $\overline{\int_R} f(X)d(X) = \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) < \underline{\int_R} f(X)d(X)$
- None of these

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A set of points of the form $X = X_0 + tU, t_1 \le t \le t_2$ is called ......

Answer (Please select your correct option)

- the neighbourhood.
- line segment.
- the closure.
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 4 of 31

For the function $f(x,y)=\dfrac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y)\to(0,0)$ along the line $y=x$ is -----

Answer (Please select your correct option)

- $\frac{1}{2}$
- $-\frac{1}{2}$
- 0
- 1

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

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MTH631 Real Analysis II

Question No : 9 of 34

A point on the graph of a function at which the tangent line is parallel to x-axis is called the ....

Answer (Please select your correct option)

- critical point ✓
- point of inflection
- point of relative extrema
- None of these

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TH631 Real Analysis II

Question No : 5 of 31

If f and g are continuous functions at the point X_0 in R^n, then $\frac{f}{g}$ is continuous at X_0 provided that ______

Answer (Please select your correct option)

- $g(X_0)=0$
- $g(X_0)\neq 0$
- $f(X_0)-g(X_0)=0$
- None of these

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \leq 0$ for all $X \neq X_0$. Then p is called ———

Answer (Please select your correct option)

- positive semidefinite
- positive definite
- negative semidefinite ✓
- negative definite

A function of the form $L(X) = f_{x_1}(X_0)x_1 + f_{x_2}(X_0)x_2 + \ldots + f_{x_n}(X_0)x_n$ is called ———

Answer (Please select your correct option)

- quadractic function
- differential of f at X_0
- constant function
- None of these

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Question Summary : (Attempted Question)

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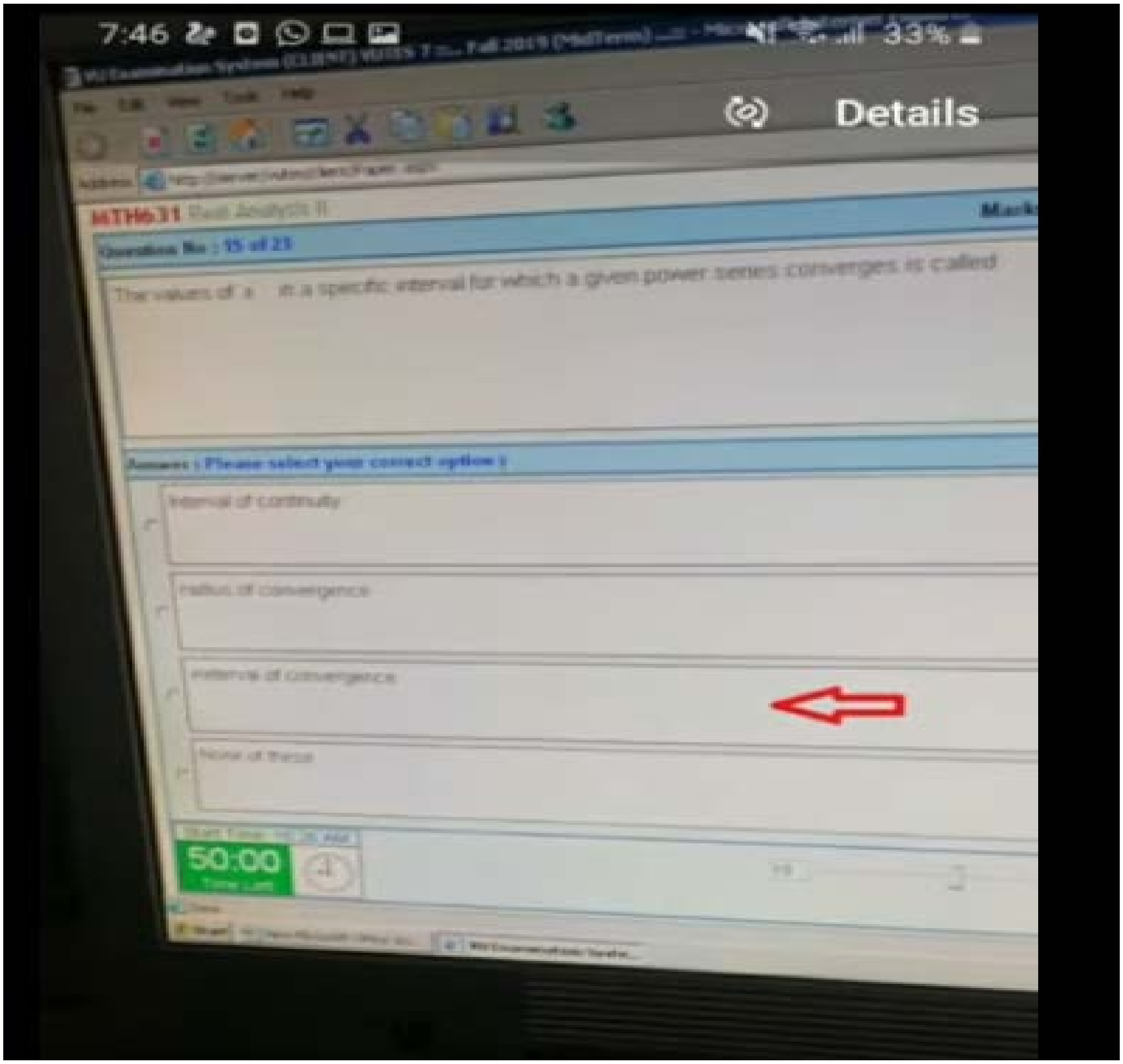
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MTH631 Real Analysis II
Question No : 15 of 23
The values of x in a specific interval for which a given power series converges is called
Answer : (Please select your correct option)
Interval of continuity
Radius of convergence
Interval of convergence
None of these
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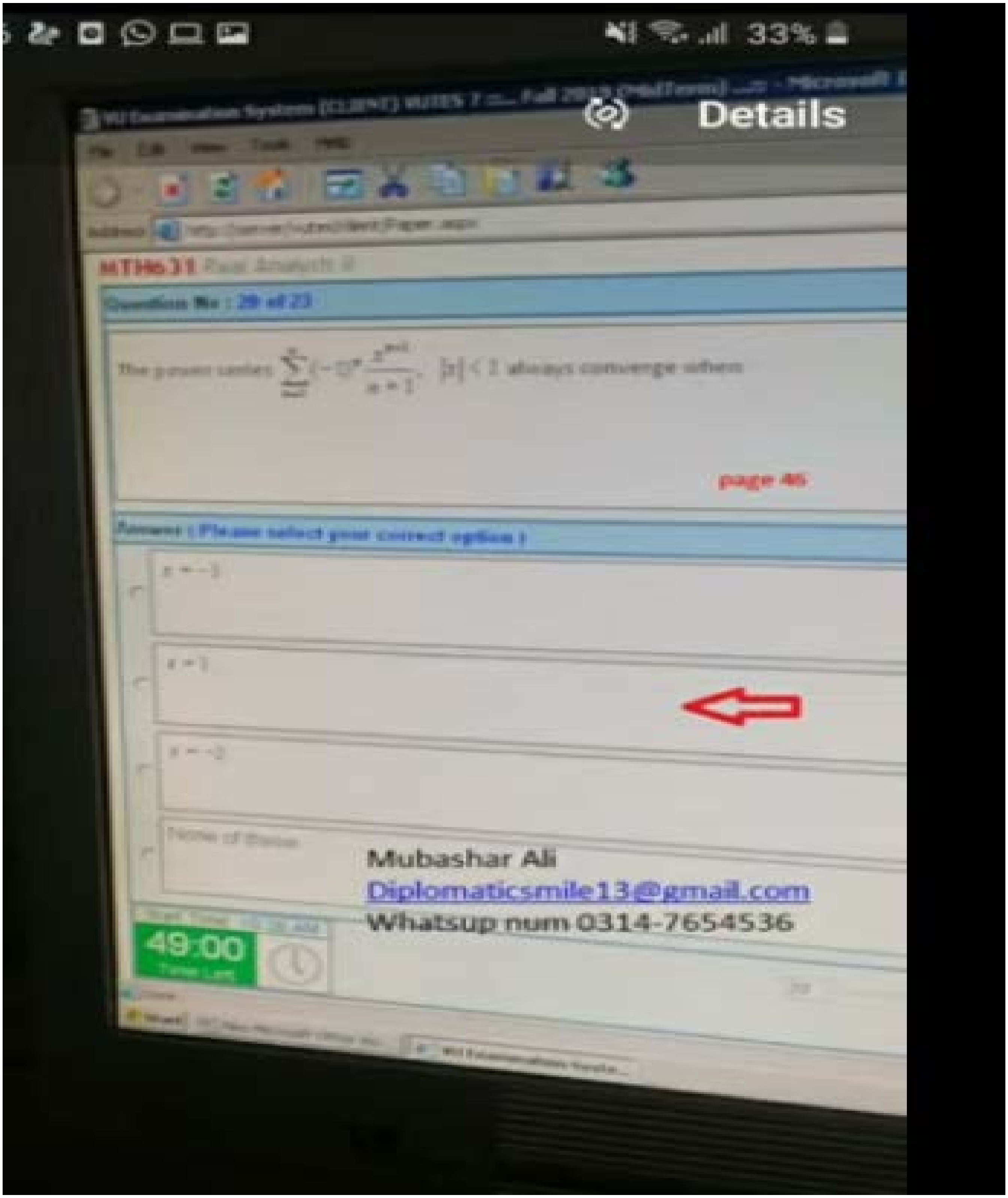
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MTH631
page 46
Answer (Please select your correct option)
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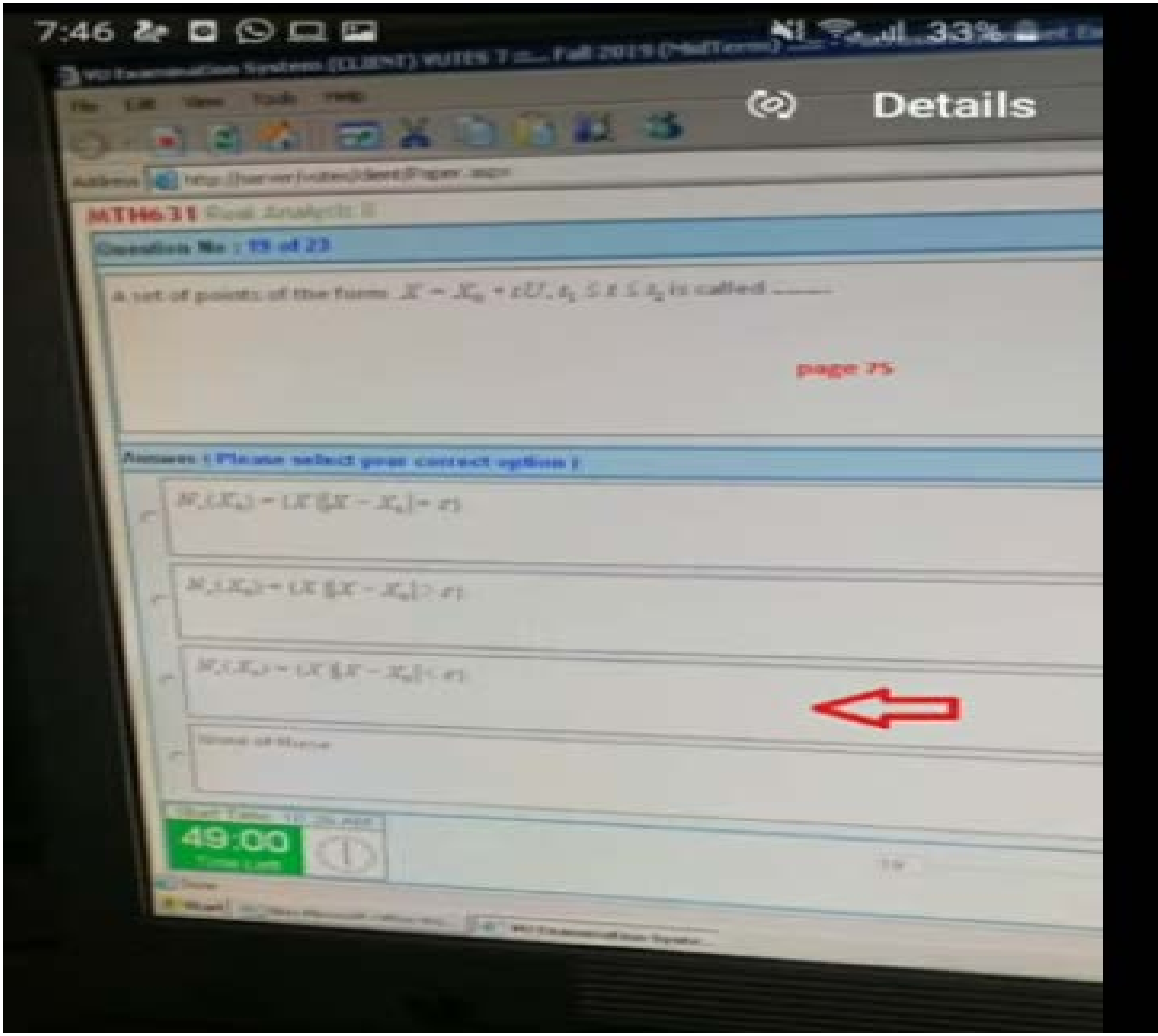

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MTH631
Question No : 19 of 23
A set of points of the form $X = X_0 + tU,\ t_1 \le t \le t_2$ is called ____
page 75
Answer (Please select your correct option)
$N_r(X_0) = \{X \mid |X - X_0| = r\}$
$N_r(X_0) = \{X \mid |X - X_0| > r\}$
$N_r(X_0) = \{X \mid |X - X_0| < r\}$
None of these
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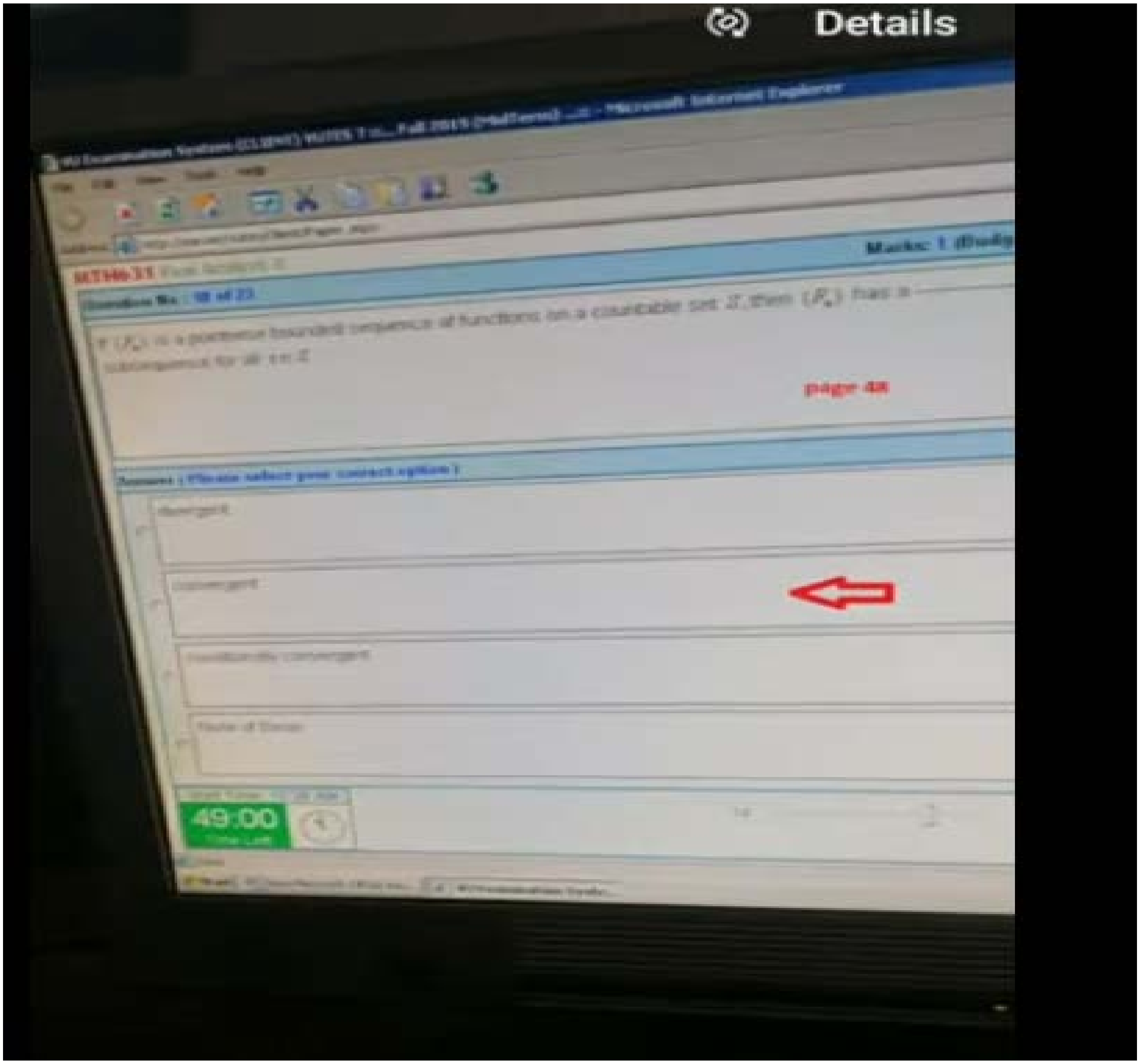
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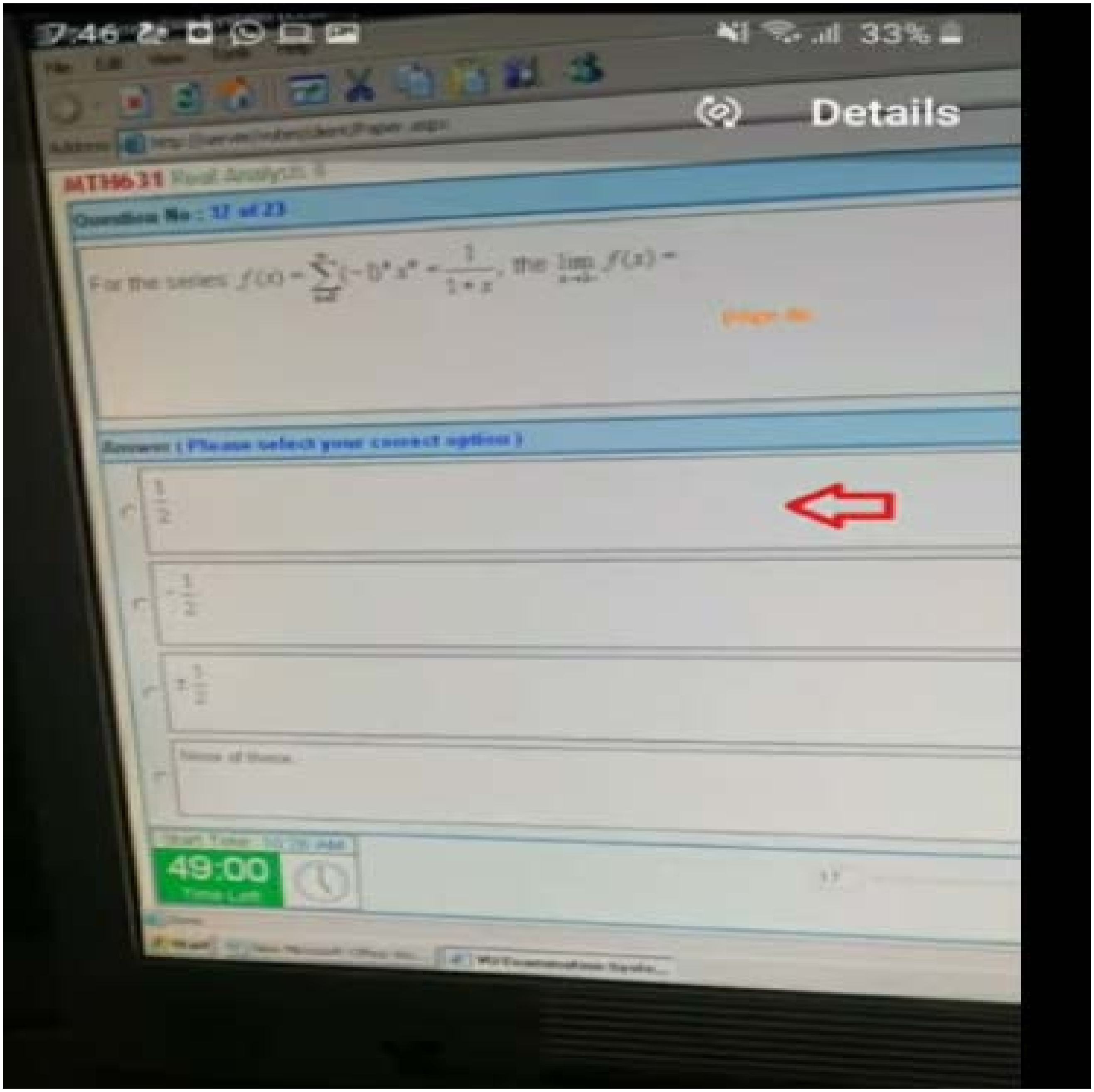
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MTH631
Question No : 17 of 23
For the series
Answer : (Please select your correct option)
None of these
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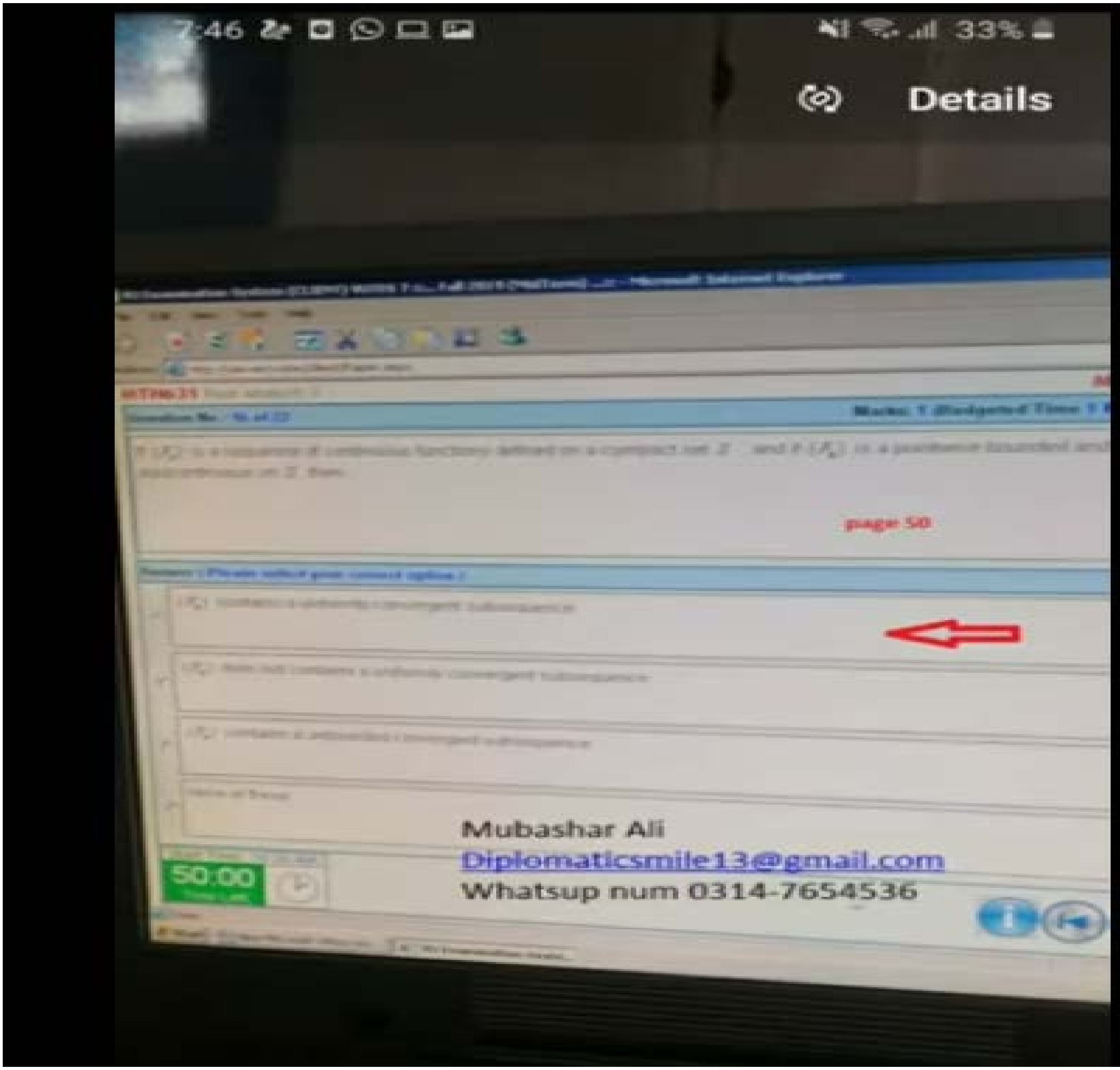
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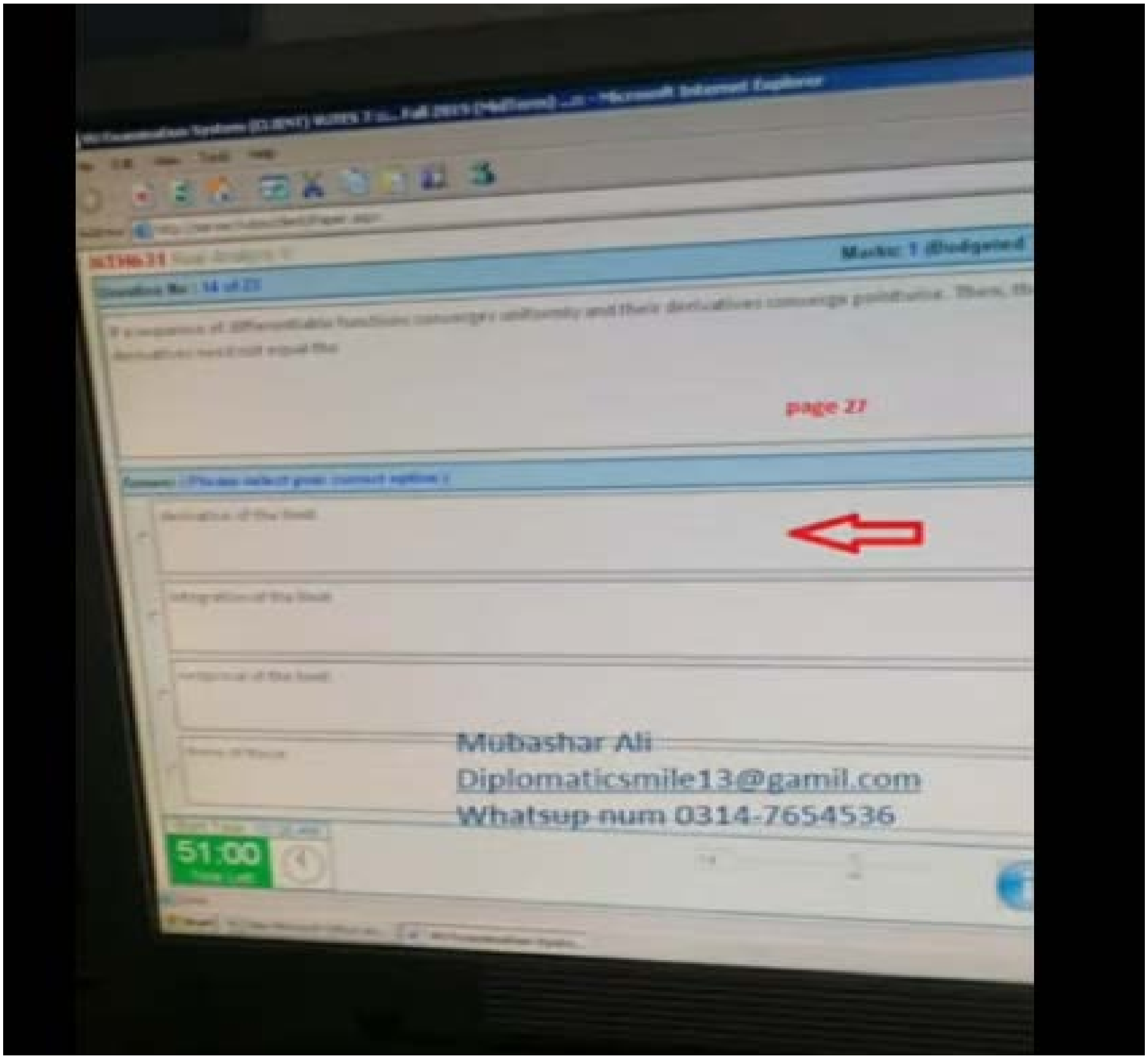
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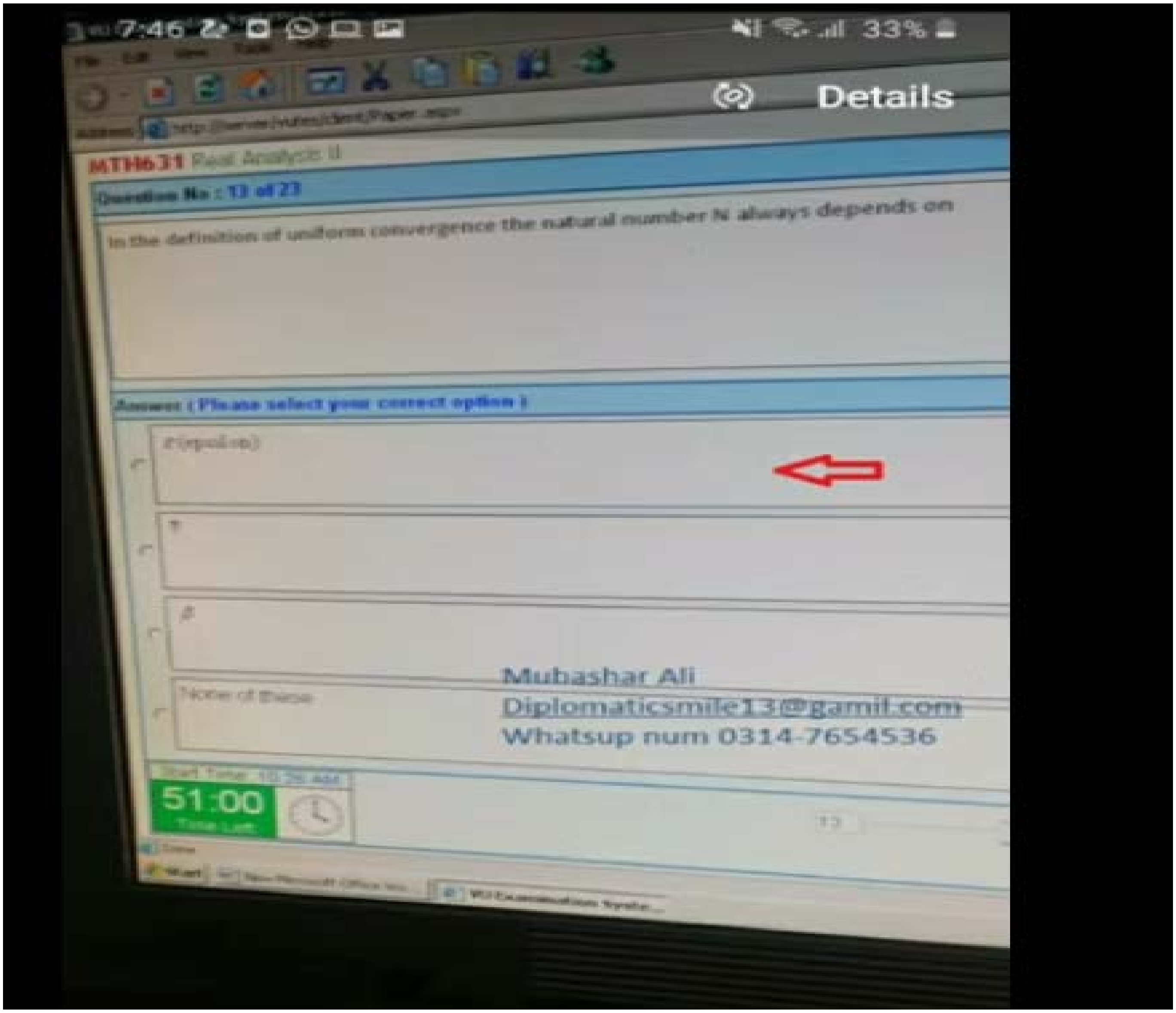
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MTH631 Real Analysis II
Question No : 13 of 23
In the definition of uniform convergence the natural number N always depends on
Answer (Please select your correct option)
None of these
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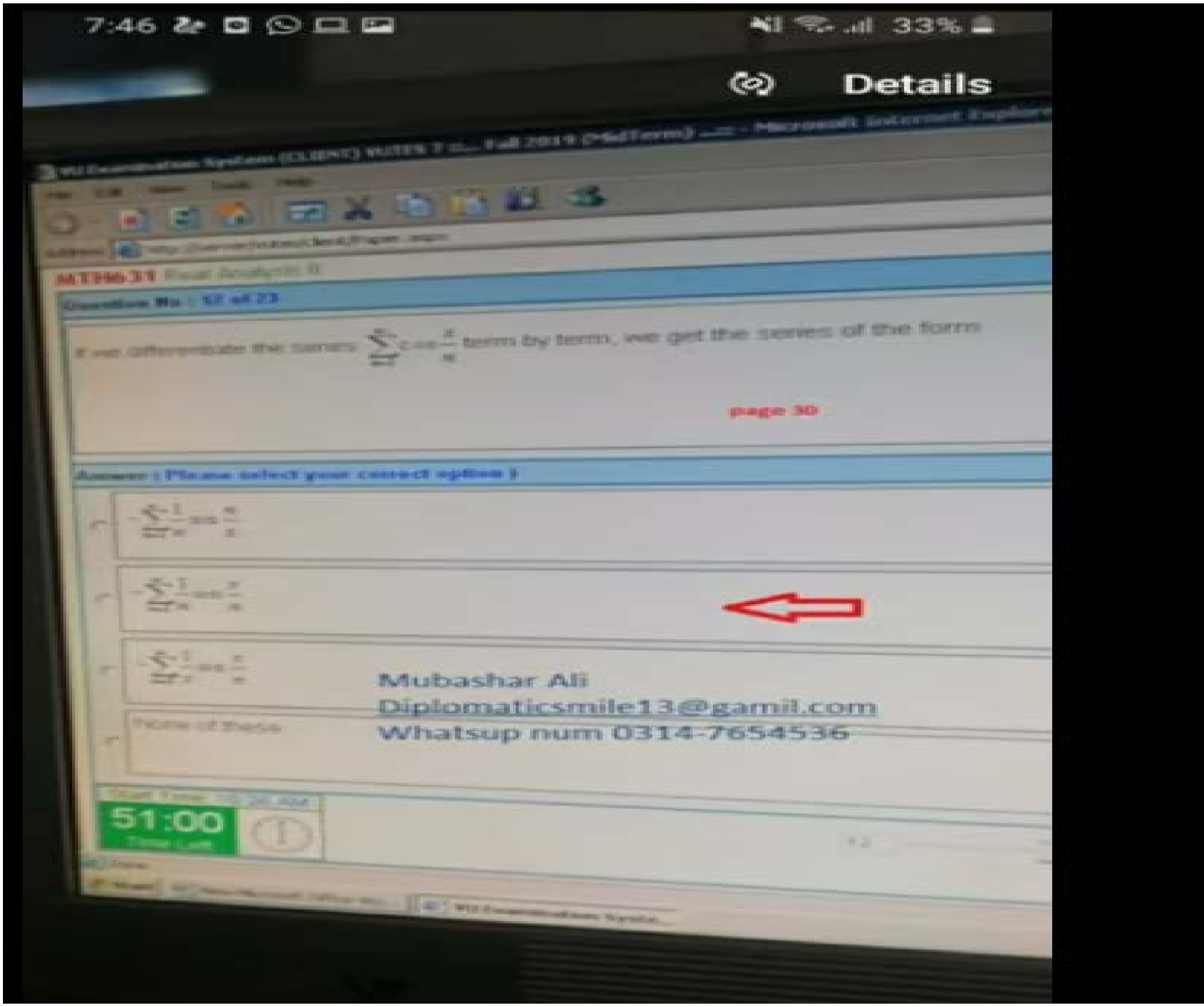
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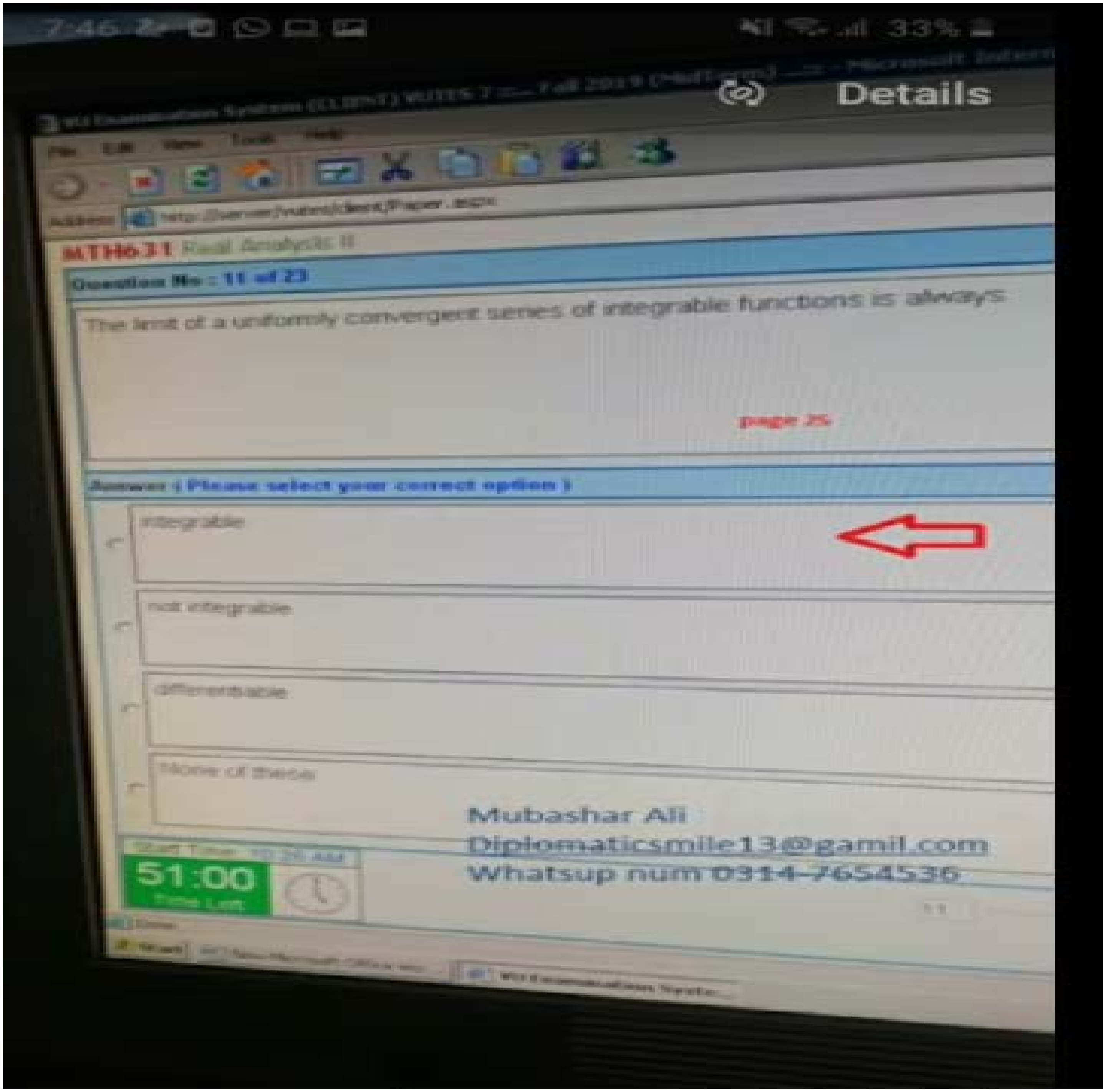
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MTH631 Real Analysis II
Question No : 11 of 23
The limit of a uniformly convergent series of integrable functions is always
page 25
Answer (Please select your correct option)
integrable
not integrable
differentiable
None of these
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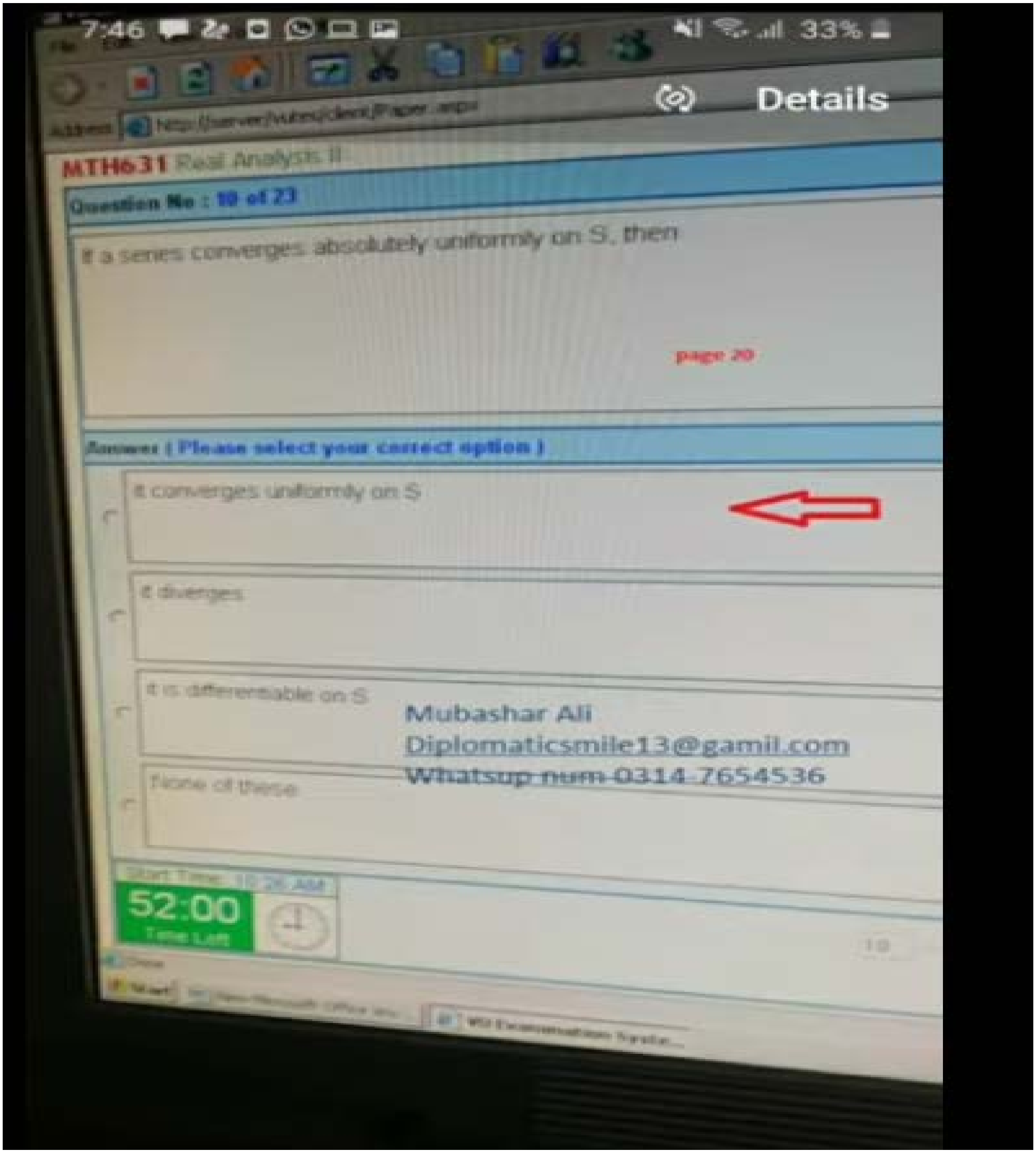
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MTH631 Real Analysis II
Question No : 10 of 23
If a series converges absolutely uniformly on S, then
page 20
Answer (Please select your correct option)
it converges uniformly on S
it diverges
it is differentiable on S
None of these
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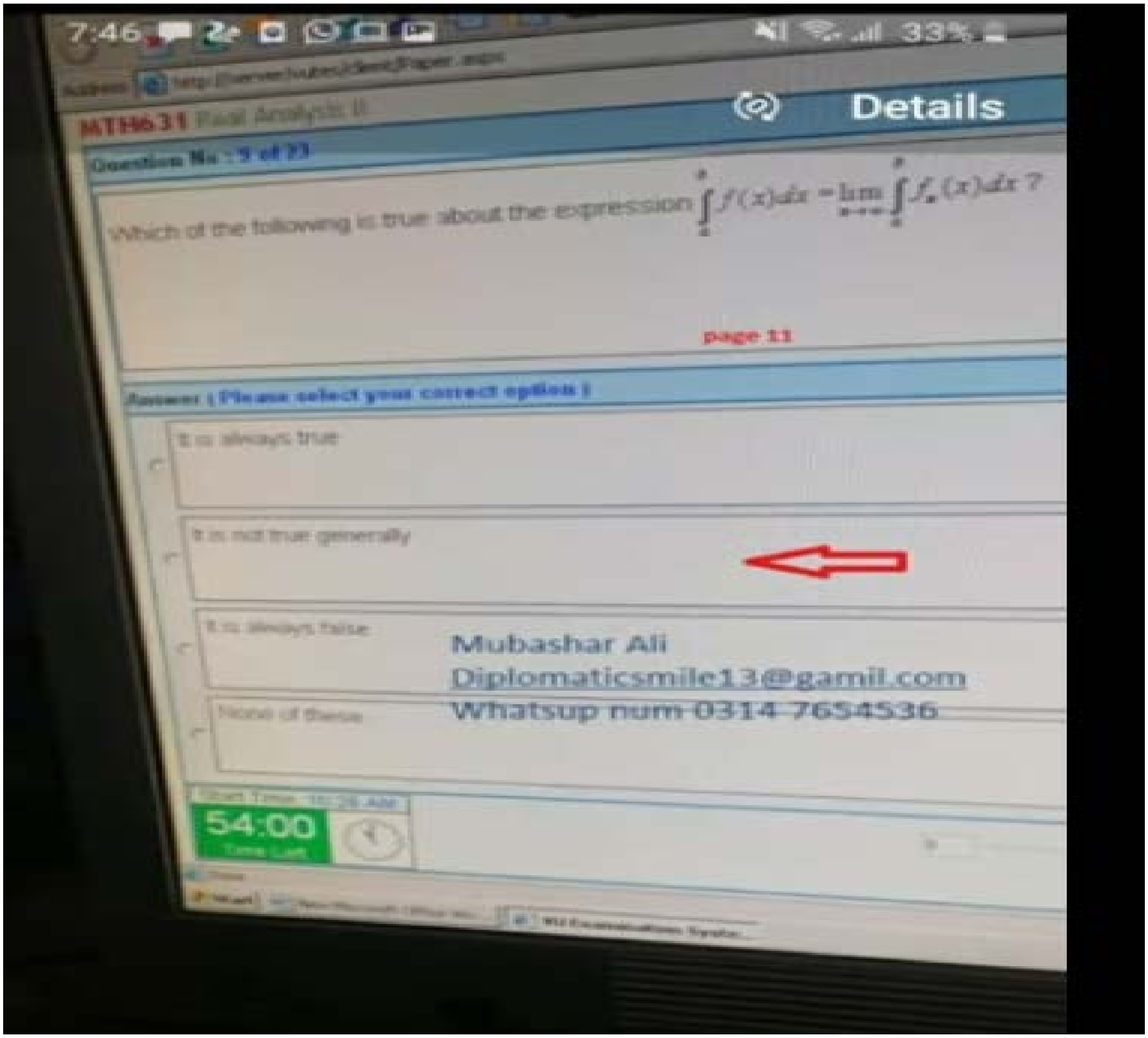
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Which of the following is true about the expression $\int_a^b f(x)dx = \lim_{n\to\infty}\int_a^b f_n(x)dx$?
page 11
It is always true
It is not true generally
It is always false
None of these
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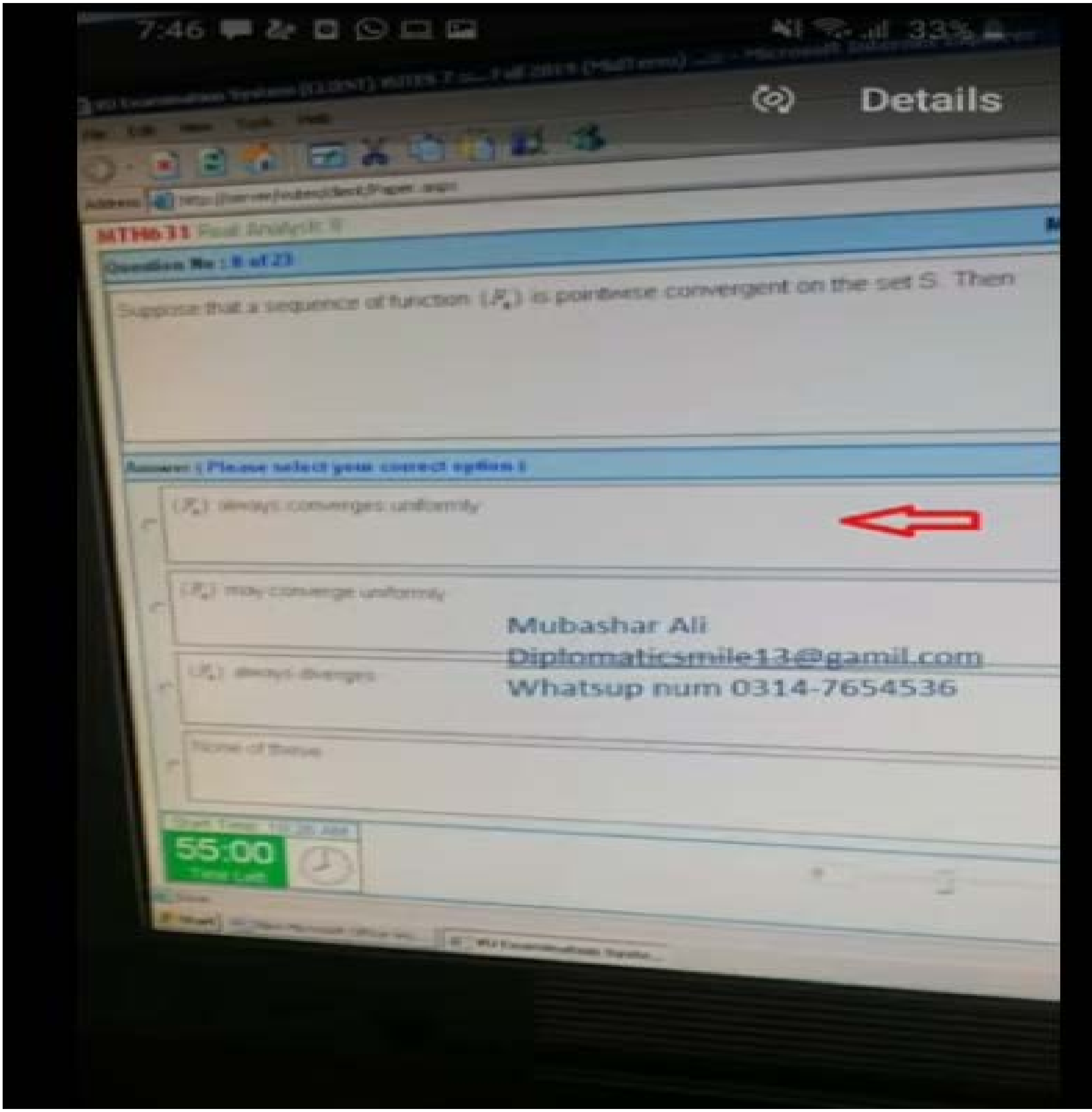
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Question No : 8 of 23
Suppose that a sequence of function (Fn) is pointwise convergent on the set S. Then
Answer (Please select your correct option)
(Fn) always converges uniformly
(Fn) may converge uniformly
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(Fn) always diverges
None of these
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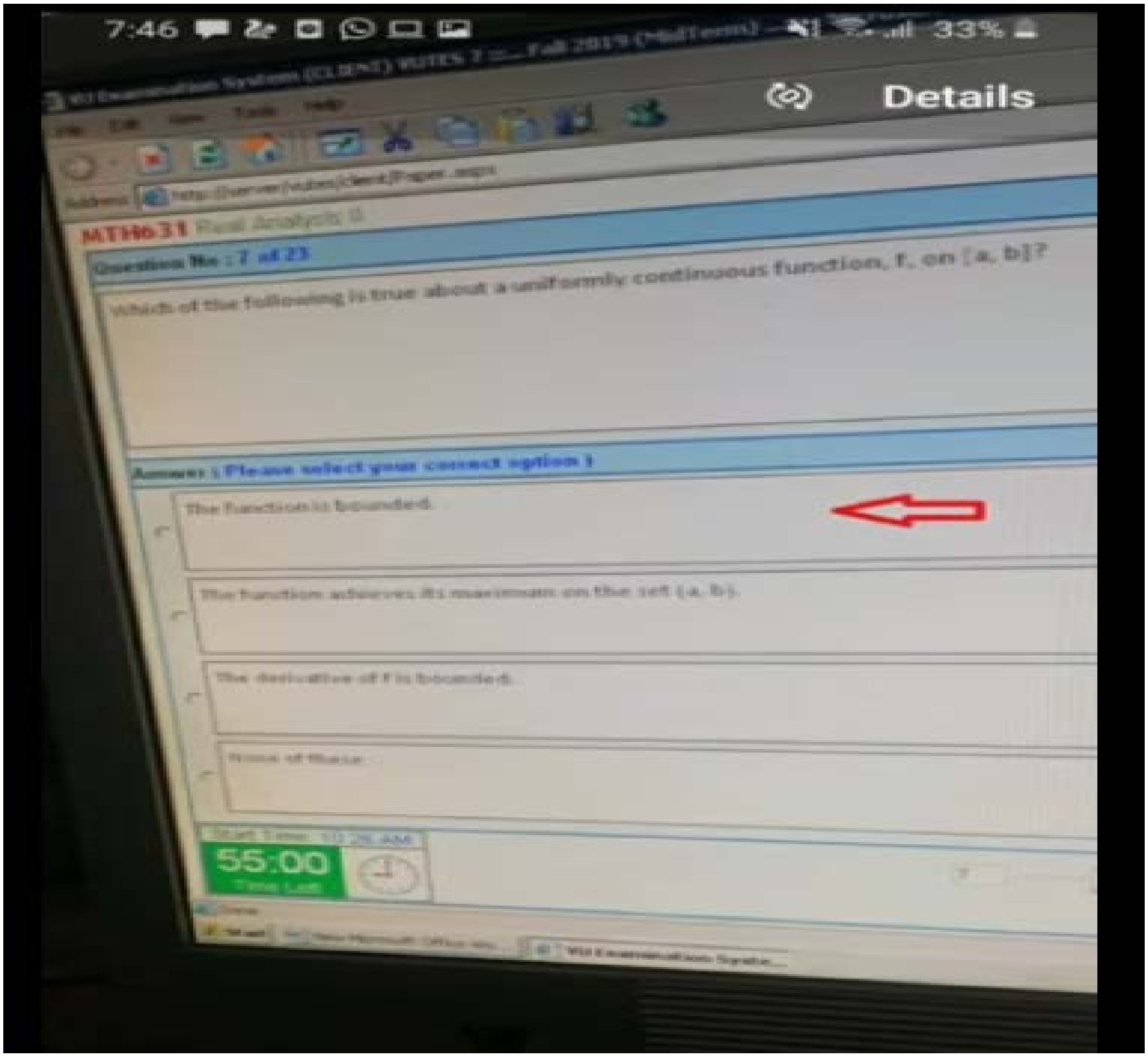
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MTH631 Real Analysis II
Question No : 7 of 23
Which of the following is true about a uniformly continuous function, f, on [a, b]?
Answer : (Please select your correct option)
The function is bounded.
The function achieves its maximum on the set (a, b).
The derivative of f is bounded.
None of these
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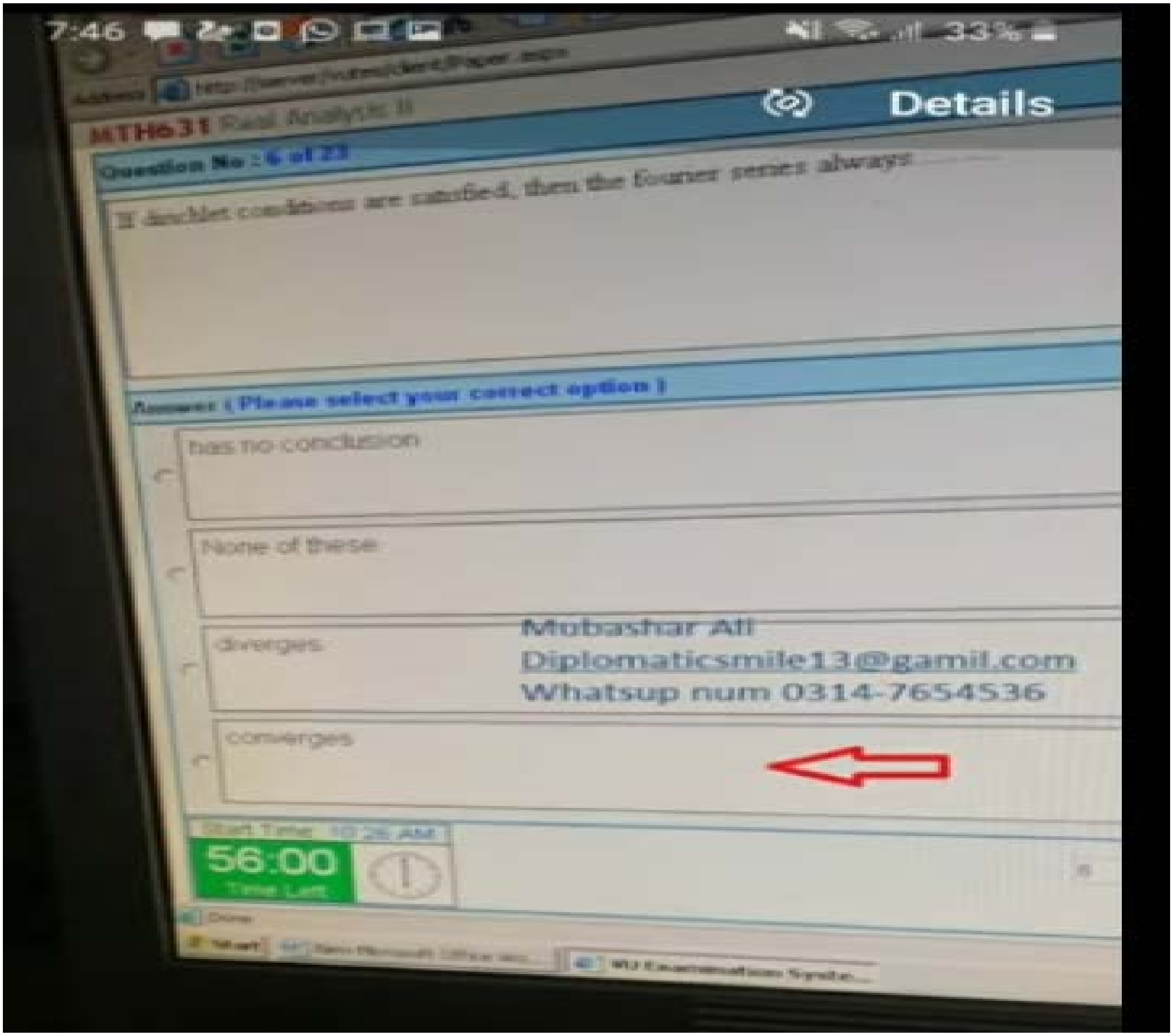
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MTH631
Question No : 6 of 23
If dirichlet conditions are satisfied, then the fourier series always
Answer: (Please select your correct option)
has no conclusion
None of these
diverges
converges
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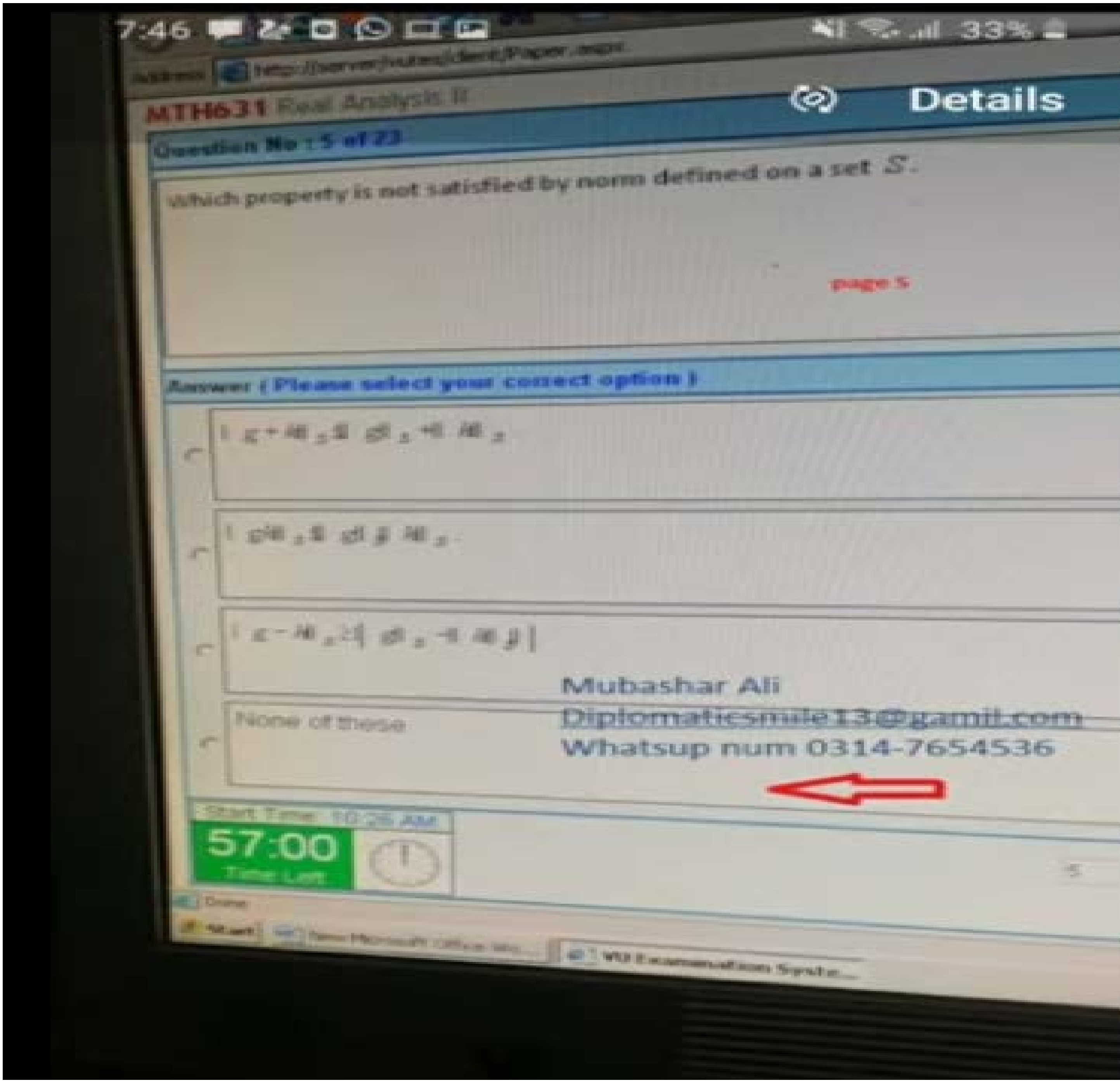
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MTH631 Real Analysis II
Question No : 5 of 23
Which property is not satisfied by norm defined on a set S.
page 5
Answer: (Please select your correct option)
None of these
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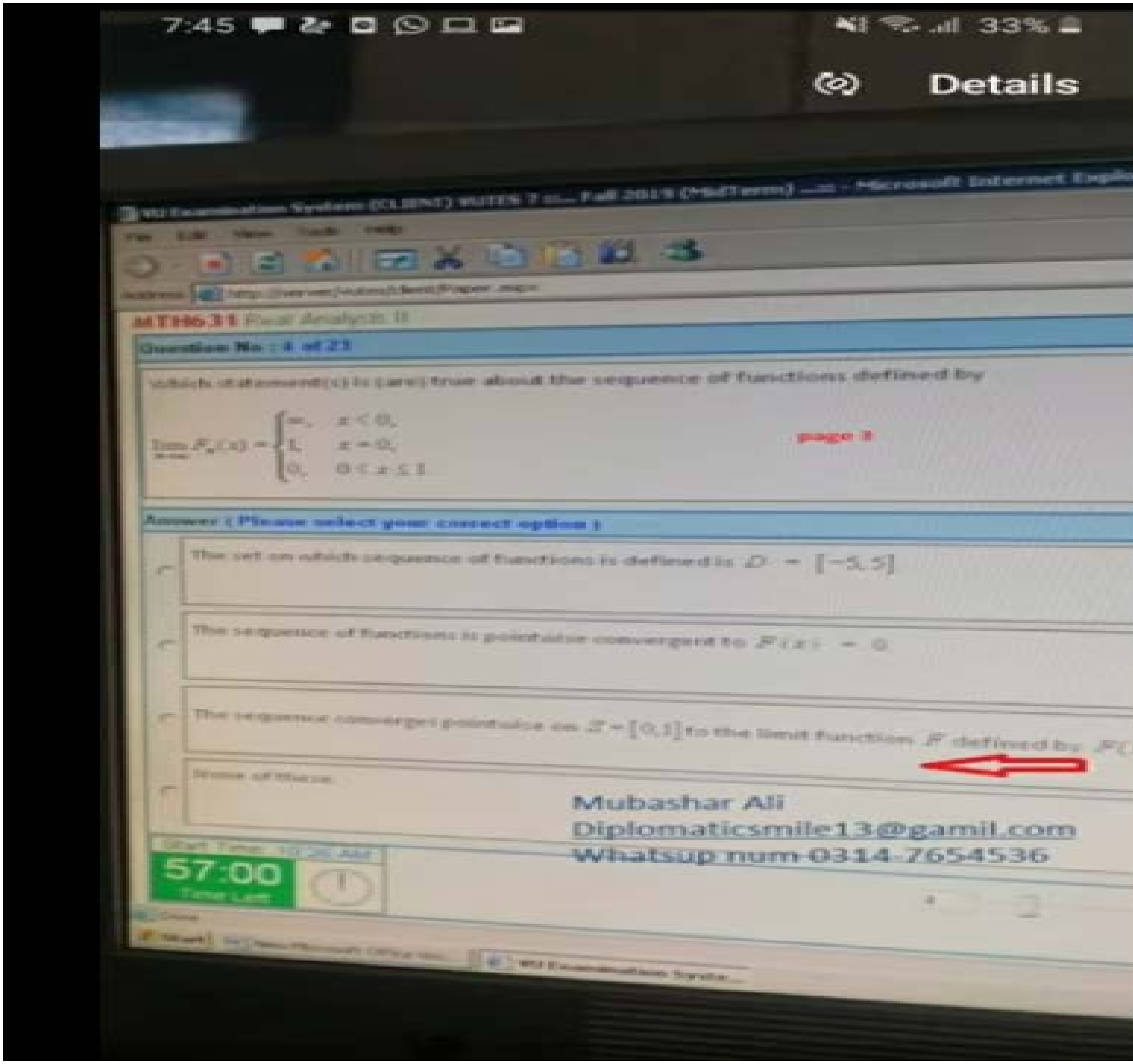
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MTH631 Real Analysis II
Question No : 4 of 23
Which statement(s) is (are) true about the sequence of functions defined by
page 3
Answer (Please select your correct option)
The set on which sequence of functions is defined is
The sequence of functions is pointwise convergent to
None of these
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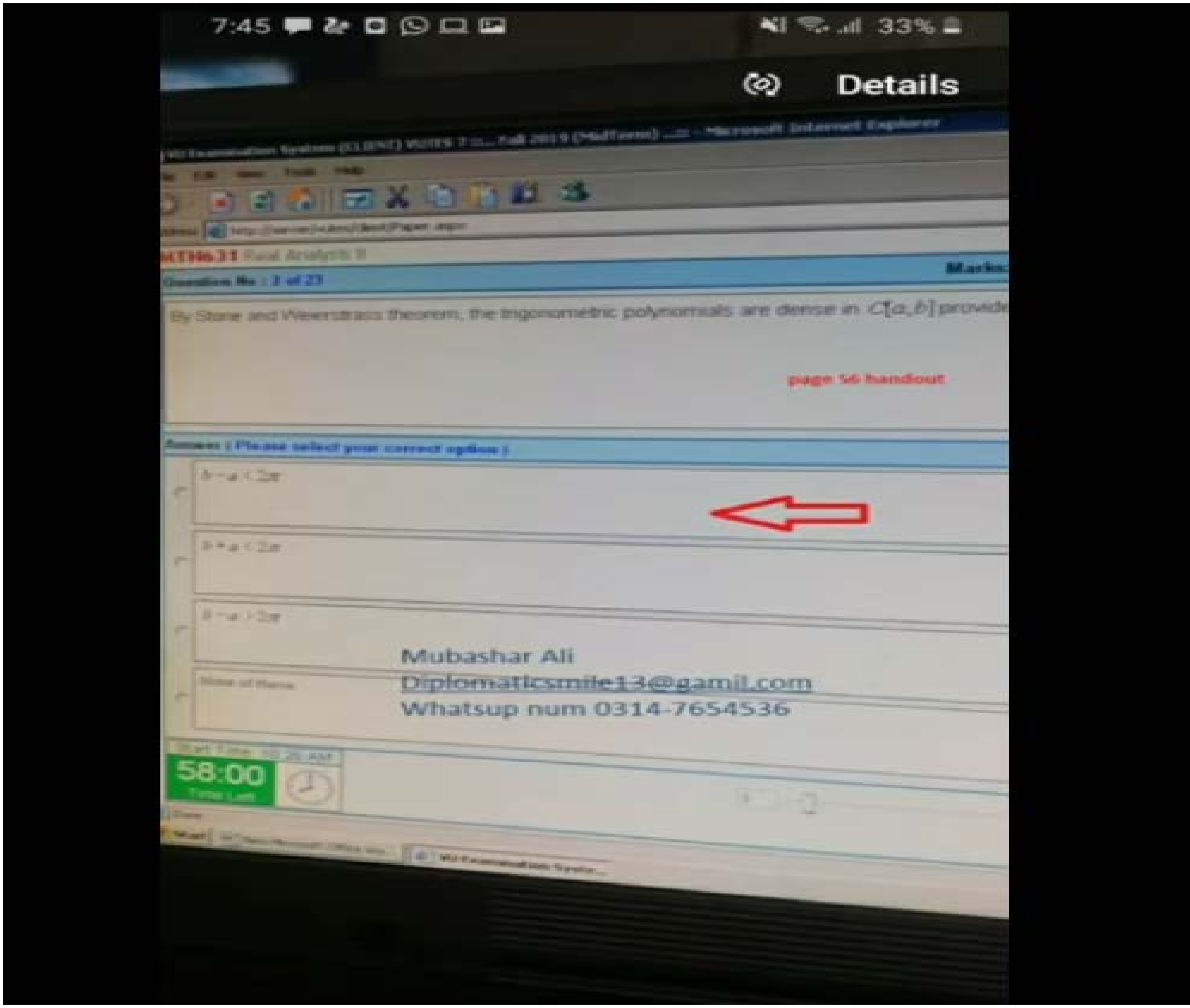
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By Stone and Weierstrass theorem, the trigonometric polynomials are dense in C[a,b] provide
page 56 handout
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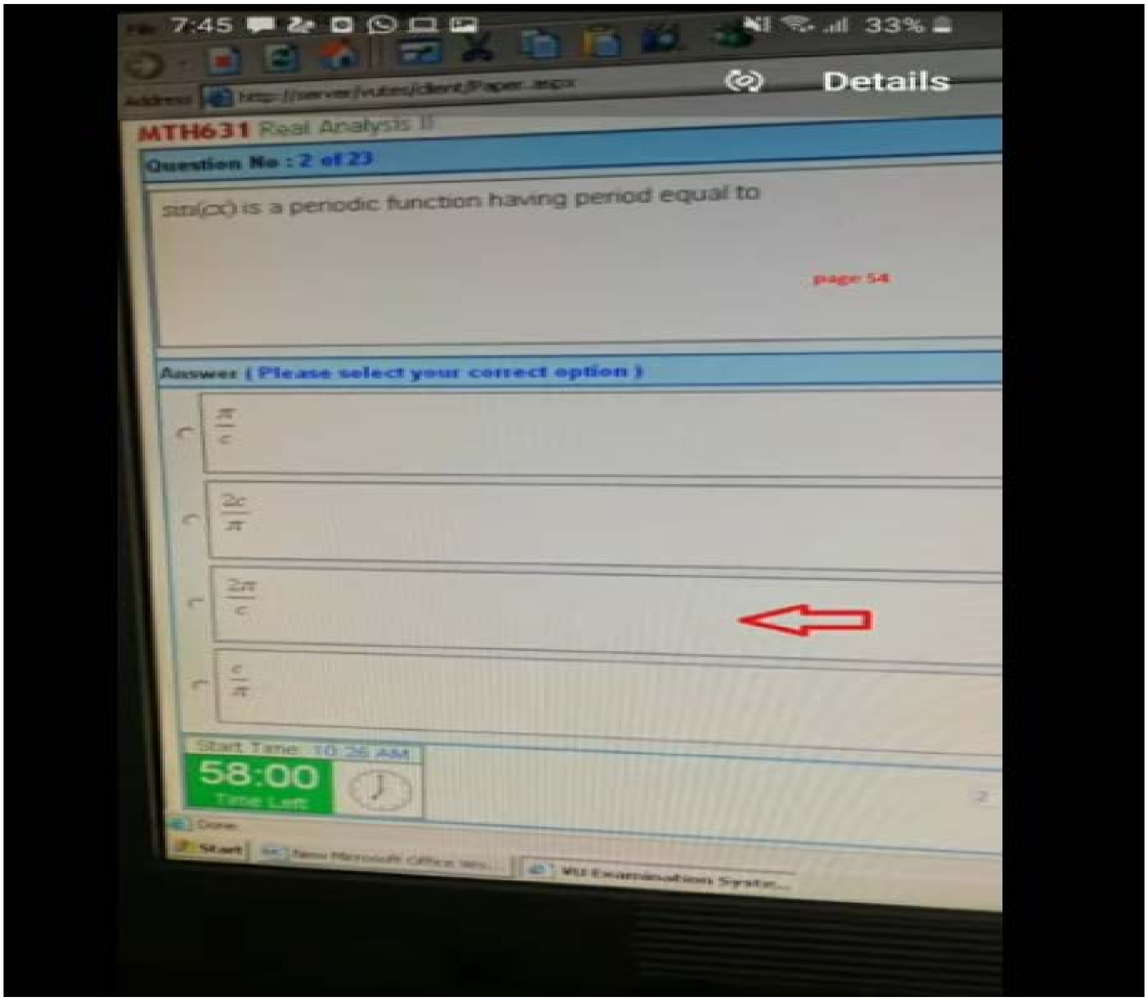
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MTH631 Real Analysis II
Question No : 2 of 23
sin(cx) is a periodic function having period equal to
page 54
Answer (Please select your correct option)
$\frac{\pi}{c}$
$\frac{2c}{\pi}$
$\frac{2\pi}{c}$
$\frac{c}{\pi}$
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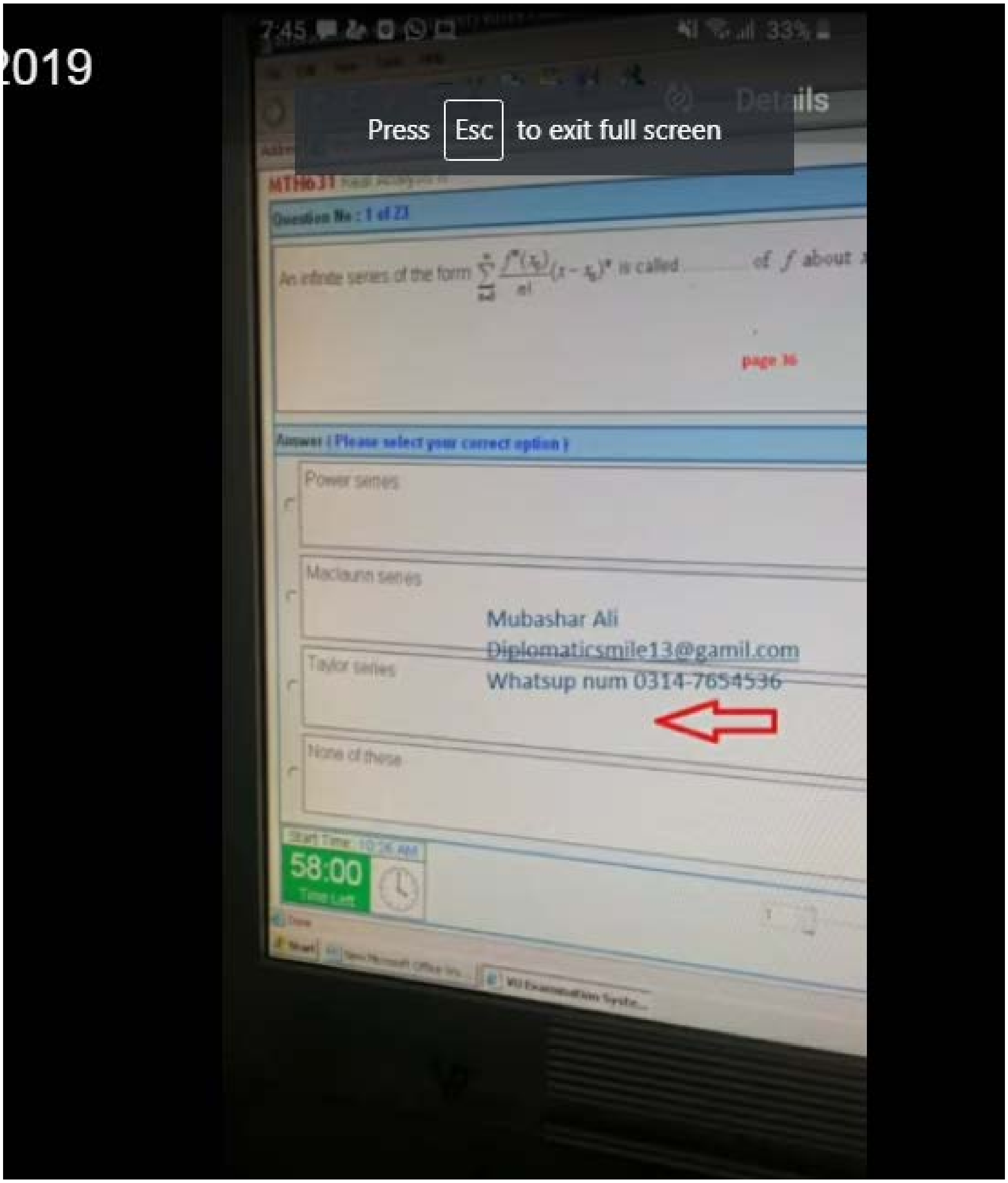
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Question No : 1 of 23
An infinite series of the form $\sum_{n=0}^{\infty} \frac{f^{n}(x_0)}{n!}(x-x_0)^n$ is called ______ of f about
Answer (Please select your correct option)
Power series
Maclaurin series
Taylor series
None of these
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MTH631 Real Analysis II

Question No : 20 of 26

If f and g are integrable on S and $f(X) \leq g(X)$ for X in S, then ———

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Answers (Please select your correct option)

- $\int_S f(X)dX \geq \int_S g(X)dX$
- $\int_S f(X)dX \leq \int_S g(X)dX$
- $\int_S f(X)dX > \int_S g(X)dX$
- None of these

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